
Exam

June 10th, 2022

Exercise 1

(42 points)

The dataset `CO2` gives the result of a controlled experiment conducted on 12 individual plants. Each of these plants was successively exposed to 7 different CO₂ concentrations while their CO₂ uptake rates were measured. Some of the plants were exposed to a cold treatment, others were not. Each of the plants, of the species *Echinochloa crus-galli*, has one of two origins: Quebec or Mississippi. The goal of the experiment is to determine if the cold treatment has an impact on the growth of plants, for which the absorption of CO₂ is a proxy.

The dataset contains 84 observations with 5 columns:

- `Plant`: a factor identifying each plant.
- `Type`: a factor with levels `Quebec` or `Mississippi` indicating the origin of the plant.
- `Treatment`: a factor with levels `nonchilled` or `chilled`, indicating if a plant was exposed to a cold treatment or not.
- `conc`: a numeric vector giving the CO₂ concentration to which the plants were exposed.
- `uptake`: a numeric vector giving the CO₂ uptake rates.

We start by naively fitting two GLM models to this data as presented in [Output 1](#) and [Output 2](#).

Output 1: GLM Model 1.

```
1 > fit_1 <- glm(uptake ~ conc + Treatment + Type, family = gaussian(), data=CO2)
2 > fit_1
3
4 Call:  glm(formula = uptake ~ conc + Treatment + Type, family = gaussian(),
5         data = CO2)
6
7 Coefficients:
8      (Intercept)          conc Treatmentchilled  TypeMississippi
9      29.25981         0.01773         -6.85952         -12.65952
10
11 Degrees of Freedom: 83 Total (i.e. Null);  80 Residual
12 Null Deviance:      9707
13 Residual Deviance: 3068      AIC: .....
14
15 > logLik(fit_1)
16 'log Lik.' -270.3099 (df=.....)
17
18 > sigma(fit_1)
19 [1] 6.193074
```

Output 2: GLM Model 2

```
1 > fit_2 <- glm(log(uptake) ~ log(conc-92) + Treatment + Type, family = gaussian(), data=CO2)
2 > fit_2
3
4
5 Call:  glm(formula = log(uptake) ~ log(conc - 92) + Treatment + Type,
6       family = gaussian(), data = CO2)
7
8 Coefficients:
9      (Intercept)      log(conc - 92)  Treatmentchilled  TypeMississippi
10          2.7085           0.1774          -0.2993           -0.4896
11
12 Degrees of Freedom: 83 Total (i.e. Null);  80 Residual
13 Null Deviance:      17.69
14 Residual Deviance: 2.459      AIC: -48.22
15
16 > sigma(fit_2)
17 [1] 0.1753334
```

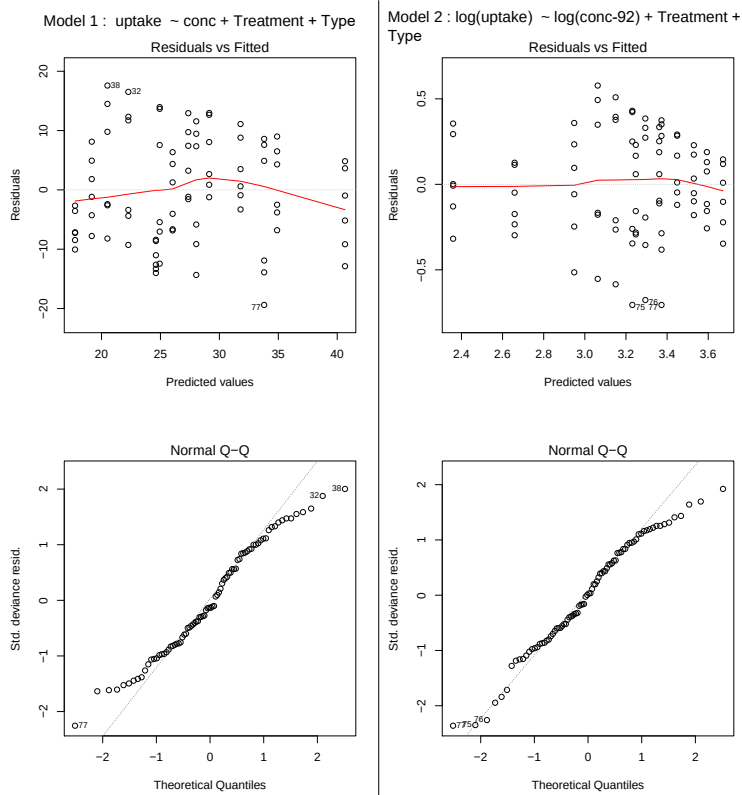
1. (1 point) Model 1 and Model 2 have been fitted using the function `glm`. Which other R function could have been used instead for this case?

Solution: Function `lm`, because Gaussian family.

2. (2 points) Based on the diagnostic plots of Figure 1, which of Model 1 or Model 2 seems to be the more relevant?

Solution: GLM Model 2 seems better: its residuals are closer to normality than the residuals of GLM Model 1, and the trend in the "residuals vs fitted values" plot is less pronounced.

Figure 1: Diagnostic plots of the fits of the Model 1 (two left hand side panels) and Model 2 (two right hand side panels)



3. (3 points) For Model 1 and Model 2 give the value of the deviance D and of the scaled deviance D^* . Perform a goodness-of-fit test at 5% level to test whether the model is good.

Solution: For Model 1, from Output 1, $D = 3068$ and $\hat{\phi} = \hat{\sigma}^2 = 6.19^2$, therefore $D^* = 3068/6.19^2 = 80.07$.

For Model 2, from Output 2, $D = 2.459$ and $\hat{\phi} = \hat{\sigma}^2 = 0.175^2$, therefore $D^* = 2.459/0.175^2 = 80.29$.

The 95% quantile of a χ_{79}^2 distribution is 100.75 (and the 95% quantile of a χ_{80}^2 distribution is 101.88), therefore the null hypothesis H_0 : “the model is good” is not rejected nor for Model 1 nor for Model 2.

4. (2 points) Compute the Akaike Information Criterion (AIC) of Model 1.

Solution:

$AIC = -2 \loglik + 2 * (\text{nb of parameters}) = -2 * (-270.3099) + 2 * 5 = 550.6198$.

We also accept the solution $AIC = -2 * (-270.3099) + 2 * 4 = 548.6198$

5. (2 points) Why can't you naively compare the AIC of Model 1 and 2 ? Explain briefly, without doing it, how you could compute the AIC of the second model in order to compare the two models.

Solution: We can't naively compare the AIC of Model 1 and 2 because the models don't have the same *response*.

The second model actually models $\log(\text{uptake})$ rather than uptake, i.e. it assumes that uptake follows a log-normal distribution. AIC of Model 2 can be computed by hand. Number of parameters is $k = 4 + 1$ and the likelihood can be computed as

$$l = \sum_i \log \left(\frac{1}{\text{uptake}_i \hat{\sigma}} \varphi \left(\frac{\log(\text{uptake}_i) - \hat{y}_i}{\hat{\sigma}} \right) \right).$$

where φ is the density of a standard normal, using the fact that the density of a standard log-normal random variable is $\frac{1}{x} \varphi(x)$. The AIC is then $2(k - l)$.

6. (2 points) Figure 2 shows, for Model 2, the residuals as a function of the randomized residuals. Explain what you observe and why.

Solution: Randomized residuals and residuals are proportional. This is expected since we are using a gaussian family. Randomized residuals are thus $\Phi(F(y_i; \hat{\mu}_i, \hat{\sigma})) = \Phi^{-1}(\Phi((y_i - \hat{y}_i)/\hat{\sigma})) = r/\hat{\sigma}$ where r are the model's residuals.

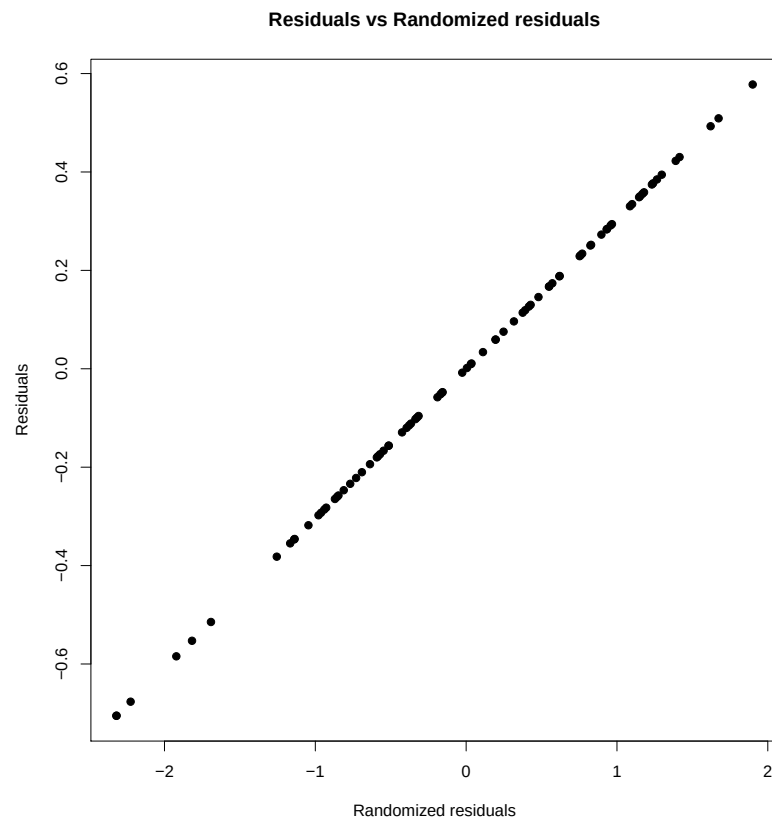


Figure 2: Residuals vs randomized residuals of Model 2.

7. (2 points) We fitted a third GLM model, Model 3 in Output 3. What is wrong with Model 3? Why are some coefficients estimated as 'NA'?

Solution: The variable Plant is nested in the variables Treatment and Type, hence the model is non-identifiable. This is why the coefficients for Type and Treatment are NA .

Output 3: GLM Model 3.

```

1 > fit_3 <- glm(log(uptake) ~ log(conc-92) + Plant + Treatment + Type , family = gaussian(),
2   data=CO2)
3 > summary(fit_3)
4
5 Call:
6 glm(formula = log(uptake) ~ log(conc - 92) + Plant + Treatment +
7   Type, family = gaussian(), data = CO2)
8
9 Deviance Residuals:
10      Min       1Q   Median       3Q      Max
11 -0.46623  -0.06363  -0.00039   0.06051   0.26150
12
13 Coefficients: (2 not defined because of singularities)
14              Estimate Std. Error t value Pr(>|t|)
15 (Intercept)      2.314026    0.039827  58.101 < 2e-16 ***
16 log(conc - 92)    0.177382    0.007446  23.823 < 2e-16 ***
17 Plant.1          -0.911548    0.045756 -19.922 < 2e-16 ***
18 Plant.2          -0.199034    0.045756  -4.350 4.47e-05 ***
19 Plant.3           0.192827    0.045756   4.214 7.26e-05 ***
20 Plant.4           0.177825    0.045756   3.886 0.000226 ***
21 Plant.5           0.192450    0.045756   4.206 7.47e-05 ***
22 Plant.6           0.002603    0.045756   0.057 0.954785
23 Plant.7          -0.168129    0.045756  -3.674 0.000459 ***
24 Plant.8          -0.232960    0.045756  -5.091 2.80e-06 ***
25 Plant.9          -0.359907    0.045756  -7.866 2.96e-11 ***
26 Plant.10         -0.019746    0.045756  -0.432 0.667381
27 Plant.11          0.040542    0.045756   0.886 0.378577
28 Treatmentchilled      NA         NA      NA      NA
29 TypeMississippi       NA         NA      NA      NA
30 ---
31 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
32
33 (Dispersion parameter for gaussian family taken to be 0.0146551)
34
35 Null deviance: 17.6913 on 83 degrees of freedom
36 Residual deviance: 1.0405 on 71 degrees of freedom
37 AIC: -102.47
38
39 Number of Fisher Scoring iterations: 2

```

8. (2 points) We could fit an alternative model as in Output 4.

Output 4: GLM Model 4.

```

1 fit_4 <- glm(uptake ~ log(conc-92) + Treatment + Type, family = gaussian(link="log"),
2   data=CO2)

```

Is Model 4 the same as Model 2? Explain in details.

Solution: The two models are not the same. Model 2 is $E(\log(Y_i)) = x_i^T \beta$, whereas Model 4 is $\log(E(Y_i)) = x_i^T \beta$. They are not equivalent.

We went back to the data description and realized that the GLM model is probably not appropriate given the structure of the data, see Figure 3.

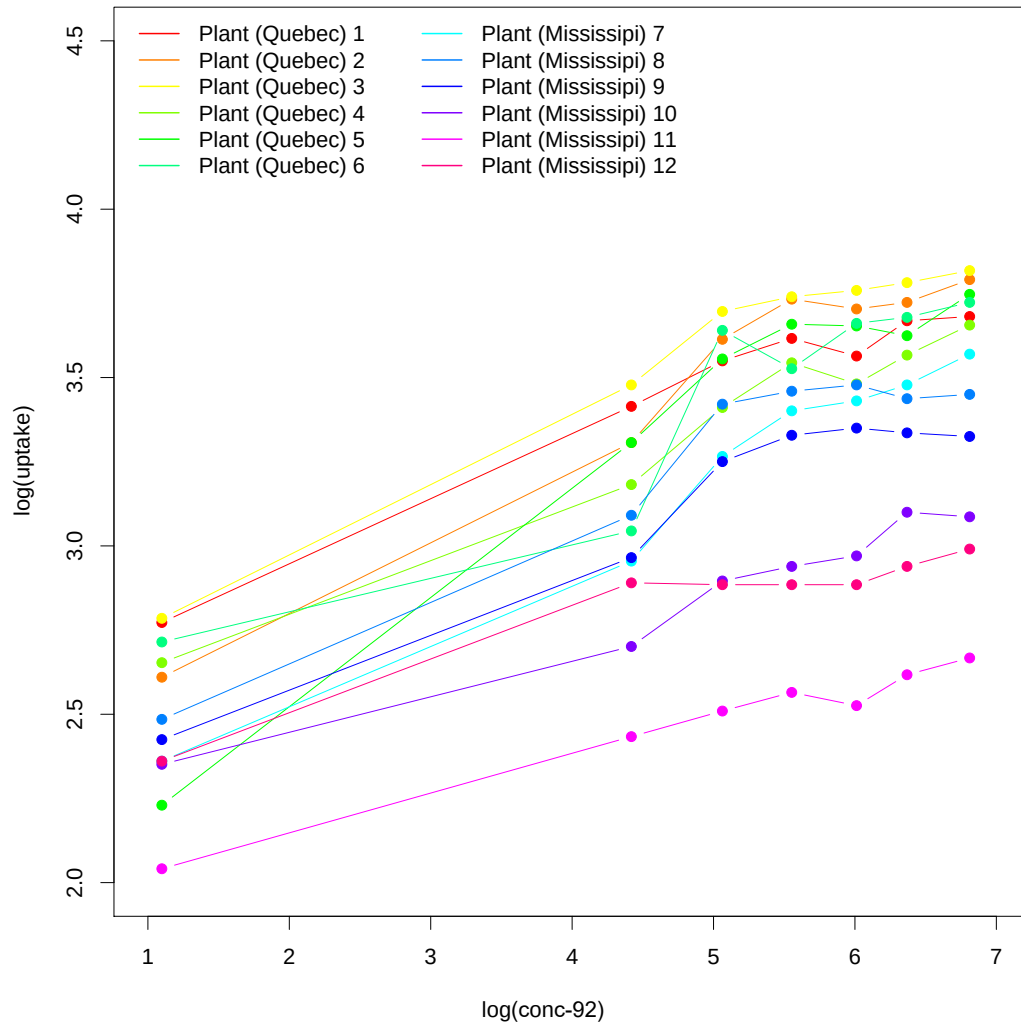


Figure 3: $\log(\text{uptake})$ versus $\log(\text{conc} - 92)$ of CO₂ for each of the 12 plants.

9. (2 points) Is it reasonable to assume that observations coming from a same plant are independent as in the GLM models? If not, give two ways to address this issue.

Solution: No. It is clear from Figure 3 that observations coming from a same plant are not independent. We could address this issue by fitting a GEE or a GLMM.

10. (3 points) We fitted a GLMM model to the CO₂ data, see Output 5. Write the model with precise mathematical notation. Explain what is random and what is not and state the assumptions on the random quantities.

Solution: If i indexes plants, and k indexes measurements, the model is as follow :

$$\log(\text{uptake})_{ik} = \alpha + \beta_1 \log(\text{conc} - 92)_{ik} + \beta_2 \text{Treatment}_{ik} + \beta_3 \text{Type}_{ik} + \gamma_i + \varepsilon_{ik}$$

where α , β_1 , β_2 and β_3 are non-random. γ_i and ε_{ik} are random with $\gamma_i \sim \mathcal{N}(0, \sigma_\gamma^2)$, $\varepsilon_{ik} \sim \mathcal{N}(0, \sigma_e^2)$ with γ_j and ε_{ik} independent for all i, j, k . In fact, Treatment and Type are time independent and therefore $\text{Treatment}_{ik} = \text{Treatment}_i$ and $\text{Type}_{ik} = \text{Type}_i$ for all k .

Output 5: GLMM model.

```
1 > fit_glmm <- lmer(log(uptake) ~ log(conc-92) + Treatment + Type + (1 | Plant), data=CO2)
2
3 > summary(fit_glmm)
4 Linear mixed model fit by REML ['lmerMod']
5 Formula: log(uptake) ~ log(conc - 92) + Treatment + Type + (1 | Plant)
6 Data: CO2
7
8 REML criterion at convergence: -73.3
9
10 Scaled residuals:
11      Min       1Q   Median       3Q      Max
12 -3.7805 -0.4962  0.0053  0.5037  1.9031
13
14 Random effects:
15  Groups   Name      Variance Std.Dev.
16  Plant    (Intercept) 0.02043  0.1429
17  Residual              0.01466  0.1211
18 Number of obs: 84, groups: Plant, 12
19
20 Fixed effects:
21              Estimate Std. Error t value
22 (Intercept)    2.708464   0.083917  32.276
23 log(conc - 92)  0.177382   0.007446  23.823
24 Treatmentchilled -0.299299   0.086643  -3.454
25 TypeMississippi -0.489577   0.086643  -5.650
26
27 Correlation of Fixed Effects:
28              (Intr) l(-92) Trtmnt
29 log(cnc-92) -0.448
30 Trtmntchlld -0.516  0.000
31 TypeMsssspp -0.516  0.000  0.000
32
33 > logLik(fit_glmm)
34 'log Lik.' 36.66494 (df=6)
```

11. (1 point) What is the name of the term `(1 | Plant)` in the GLMM model?

Solution: Random intercept.

12. (2 points) Is this GLMM model performing better than Model 2 (Output 2)? Justify.

Solution: In addition to make more sense with the given data, this model has a way lower AIC ($-2 * 36.66494 + 2 * 6 = -61.33$) than Model 2 (AIC = -48.22): it is a better fit.

13. (2 points) The variable `Treatment` is a 2-level factor. Only one coefficient is estimated. Explain why.

Solution: One of the two levels of `Treatment` is a reference level and doesn't need a coefficient since the effect of that level is already taken into account by the intercept.

14. (2 points) Is the effect of the variable `Treatment` significant at the 5% level ?

Solution: The t-value associated with the `Treatment` coefficient is -3.454. The 2.5% quantile of a normal distribution is -1.96. The variable `Treatment` is therefore significant at the 5% level.

15. (2 points) Comment on and interpret the sign of the coefficient for `Treatment`.

Solution: The sign of the coefficient of `Treatment` is negative for the level `chilled`. It indicates that chilling a plant tends to diminish its CO₂ uptake, which is expected since cold slows down plant growth.

16. (2 points) Based on the Output 5 and given the predictions of the random effects in Output 6, predict the value of uptake for the Plant `Qn3`, from Quebec, for which $\log(\text{conc} - 92) = 3$ under a chilled treatment.

Output 6: Random effect predictions of the GLMM model.

```

1 > ranef(fit_glmm)
2 $Plant
3   (Intercept)
4 Qn1 -0.12415774
5 Qn2 -0.09634598
6 Qn3 -0.02163654
7 Qc1  0.04720421
8 Qc3  0.11129942
9 Qc2  0.08363664
10 Mn3  0.02355423
11 Mn2  0.13265581
12 Mn1  0.08593023
13 Mc2 -0.30362231
14 Mc3  0.01718117
15 Mc1  0.04430088

```

Solution: Using Output 5 and 6: $\log(\text{uptake}) = 2.71 + 0.177 \times 3 - 0.299 + 0 - 0.022 = 2.92$ and therefore $\text{uptake} = \exp(2.92) = 18.54$.

17. (2 points) For some types of GLMMs a numerical procedure is required. Explain in which cases and why it is needed. Name two such numerical approximation procedures.

Solution: Except for the Gaussian family, the marginal likelihood is an integral that cannot be solved analytically. The numerical options to solve it include Laplace approximation, (adaptive) Gaussian quadrature, Monte Carlo approximation.

We fitted a GAMM (generalized additive mixed model) with a random intercept to account for possible non linearity in the relationship between $\log(\text{uptake})$ and $\log(\text{conc} - 92)$, see Output 7 and Figure 4.

Output 7: GAMM Model.

```

1 > fit_gamm <- gamm4(log(uptake) ~ s(log(conc-92), k = 7) + Treatment + Type + (1|Plant),
2   data = CO2)
3
4 > summary(fit_gamm)
5 Family: gaussian
6 Link function: identity
7
8 Formula:
9 log(uptake) ~ s(log(conc-92), k = 7) + Treatment + Type + (1|Plant)
10
11 Linear mixed model fit by REML ['lmerMod']
12 REML criterion at convergence: -81.6
13
14 Scaled residuals:
15 Min      1Q  Median      3Q      Max
16 -3.9470 -0.4795  0.0591  0.4419  2.4111
17
18 Random effects:
19 Groups   Name                Variance Std.Dev.
20 Plant    (Intercept)  0.02081   0.1443

```

```

21 Residual          0.01196  0.1094
22 Number of obs: 84, groups: Plant, 12; Xr, 5
23
24 Fixed effects:
25 Estimate Std. Error t value
26 X(Intercept)      3.60358    0.07504  48.025
27 XTreatmentchilled -0.29930    0.08664  -3.454
28 XTypeMississippi  -0.48958    0.08664  -5.650
29 Xs(log(conc-92))F1  0.19013    0.07723   2.462
30
31 Correlation of Fixed Effects:
32 X(Int) XTrtmn XTypMs
33 XTrtmntchl1 -0.577
34 XTypMssssp -0.577  0.000
35 Xs(lgcnc)F1  0.000  0.000  0.000
36
37 AIC: -67.6035
38
39 Approximate significance of smooth terms:
40          edf   Ref.df   F      p-value
41 s(log(conc-92)) 3.524   3.524  202.3 <2e-16 ***
42
43 Estimated degrees of freedom: 3.52 total = 6.52
44 R-sq.(adj) =  0.864
45 lmer.REML = -81.603 Scale est. = 0.01196 n = 84
46
47 > gam.check(fit_gamm)
48 Basis dimension (k) checking results. Low p-value (k-index<1) may
49 indicate that k is too low, especially if edf is close to k'.
50
51          k'   edf    k-index p-value
52 s(log(conc-92)) 6.00 3.52    0.39 <2e-16 ***
53 ---

```

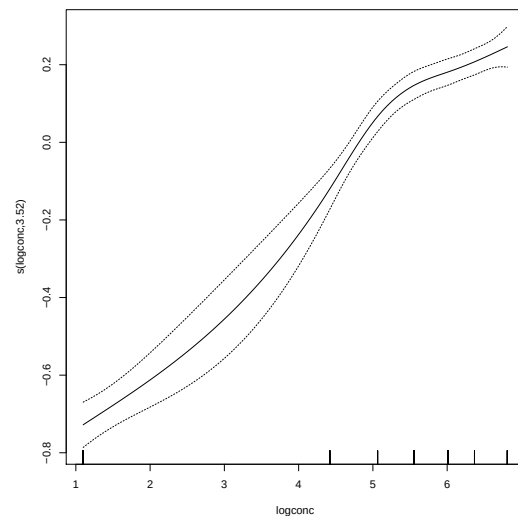


Figure 4: Fitted $s(\log(\text{conc}-92))$ term in the GAMM model.

18. (1 point) Can we compare the AIC of the GAMM model with the AIC of the GLMM model ?

Explain.

Solution: Yes, the AICs can be compared: the GLMM model of Output 5 is nested in the GAMM model of Output 7. (a linear fit is nested in a spline fit)

19. (1 point) Does the GAMM model improve over the GLMM Model ? Justify your answer.

Solution: Yes, the AIC of the GAMM model is -67.60, lower than the AIC of the GLMM model (-61.33).

20. (1 point) Does your conclusion given at point 14. change with the GAMM model ?

Solution: No, the estimated coefficient for `Treatment` is almost the same and the conclusions do not change.

21. (2 points) Is the number of basis functions associated to $\log(\text{conc}-92)$ enough, based on Output 7? Any fit with 8 basis functions or more returns an error. Why is that ?

Solution: `gam.check()` seems to indicate that 7 basis functions are not enough, but their number cannot be increased because we see from Figure 3 that the variable $\log(\text{conc} - 92)$ only takes 7 distinct values.

22. (2 points) Is the smooth term on $\log(\text{conc} - 92)$ worth with respect to a linear fit? Justify your answer.

Solution: Yes, the smooth term is worth. The approximate significance of smooth terms test is significant, therefore there is evidence of non linearity.

Other justifications:

- AIC of the GAMM model improved over GLMM model (only difference is the $\log(\text{conc}-92)$ term.
- a straight line cannot be fitted within the confidence bands.

23. (1 point) Would adding a spline on the variable `Treatment` be relevant ?

Solution: No, because `Treatment` is a dummy variable. Smooth terms are only for continuous variables.

Exercise 2

(18 points)

For the Gaussian family with identity link (like in the previous exercise), we would like to show that the maximum likelihood estimator of the regression parameter β in a random intercept GLMM is equivalent to the estimator of β in the GEE approach with exchangeable correlation matrix, **when the variance/noise parameters are known**.

Let Y_{it} denote the response of individual i at time t , for $i = 1, \dots, n$ and $t = 1, \dots, n_i$. We assume a normal distribution for Y_{it} and we define a linear predictor $x_{it}^T \beta$, where both x_{it} and β are of dimension $(p+1) \times 1$ and x_{it} is non-random.

We first consider a random intercept GLMM specification as follows

$$Y_{it} = x_{it}^T \beta + \gamma_i + \epsilon_{it}, \quad (1)$$

where we assume $\gamma_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\gamma^2)$, $\epsilon_{it} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\epsilon^2)$, with γ_i and ϵ_{jt} independent $\forall i, j, t$. We assume that σ_γ^2 and σ_ϵ^2 are known.

Model (1) can equivalently be written as

$$Y_i = X_i \beta + u_i, \quad (2)$$

where $Y_i = (Y_{i1}, \dots, Y_{in_i})^T$, $X_i = (x_{i1}, \dots, x_{in_i})^T$ and $u_i = (u_{i1}, \dots, u_{in_i})^T$, with $u_i \sim N(0, \Omega_i)$.

Reminder: If $Z = (Z_1, \dots, Z_k) \sim N(\mu, \Sigma)$, its density is

$$f(z) = (2\pi)^{-k/2} |\Sigma|^{-1/2} \exp \left(-\frac{1}{2} (z - \mu)^T \Sigma^{-1} (z - \mu) \right),$$

where $|\Sigma|$ denotes the determinant of Σ .

1. (3 points) Show that $Cov[Y_{it}, Y_{is}]$ equals $\sigma_\gamma^2 + \sigma_\epsilon^2$ if $t = s$, and equals σ_γ^2 if $t \neq s$.

Solution:

$$\begin{aligned} Cov(Y_{it}, Y_{is}) &= Cov(x_{it}^T \beta + \gamma_i + \epsilon_{it}, x_{is}^T \beta + \gamma_i + \epsilon_{is}) \\ &= Cov(\gamma_i + \epsilon_{it}, \gamma_i + \epsilon_{is}) \\ &= Cov(\gamma_i, \gamma_i) + Cov(\gamma_i, \epsilon_{is}) + Cov(\epsilon_{it}, \gamma_i) + Cov(\epsilon_{it}, \epsilon_{is}) \\ &= \text{var}(\gamma_i) + 0 + 0 + Cov(\epsilon_{it}, \epsilon_{is}) \end{aligned}$$

As ϵ_{is} and ϵ_{it} are independent if $t \neq s$, $Cov(\epsilon_{it}, \epsilon_{is}) = \begin{cases} 0 & \text{if } t \neq s \\ \text{var}(\epsilon_{it}) & \text{if } t = s \end{cases}$ Hence,
 $Cov(Y_{it}, Y_{is}) = \sigma_\gamma^2$ if $t \neq s$ and $Cov(Y_{it}, Y_{is}) = \sigma_\gamma^2 + \sigma_\epsilon^2$ if $t = s$.

2. (2 points) Define u_i as a function of γ_i and $\epsilon_i = (\epsilon_{i1}, \dots, \epsilon_{in_i})^T$.

Solution: $Y_{it} = x_{it}^T \beta + \gamma_i + \epsilon_{it}$ is equivalent to,

$$Y = X\beta + \begin{pmatrix} \gamma_i \\ \vdots \\ \gamma_i \end{pmatrix} + \begin{pmatrix} \epsilon_{i1} \\ \vdots \\ \epsilon_{in_i} \end{pmatrix} = X_i \beta + \begin{pmatrix} \gamma_i + \epsilon_{i1} \\ \vdots \\ \gamma_i + \epsilon_{in_i} \end{pmatrix}.$$

Hence $u_i = \begin{pmatrix} \gamma_i + \epsilon_{i1} \\ \vdots \\ \gamma_i + \epsilon_{in_i} \end{pmatrix}.$

3. (2 points) Give Ω_i explicitly (by making clear what are its dimensions).

Solution: $\text{cov}(u_{it}, u_{is}) = \text{cov}(\gamma_i + \epsilon_{it}, \gamma_i + \epsilon_{is}) = \begin{cases} \sigma_\gamma^2 & \text{if } t \neq s \\ \sigma_\gamma^2 + \sigma_\epsilon^2 & \text{if } t = s \end{cases}.$ Hence

$$\Omega_i = \begin{pmatrix} \sigma_\gamma^2 + \sigma_\epsilon^2 & \sigma_\gamma^2 & \cdots & \sigma_\gamma^2 \\ \sigma_\gamma^2 & \sigma_\gamma^2 + \sigma_\epsilon^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \sigma_\gamma^2 \\ \sigma_\gamma^2 & \cdots & \sigma_\gamma^2 & \sigma_\gamma^2 + \sigma_\epsilon^2 \end{pmatrix}.$$

This matrix is of dimension $n_i \times n_i$.

4. (2 points) What is the distribution of Y_i ?

Solution: As $u_i \sim \mathcal{N}(0, \Omega_i)$ and $Y_i = X_i \beta + u_i$, $Y_i \sim \mathcal{N}(X_i \beta, \Omega_i)$.

5. (3 points) Assuming Ω_i known $\forall i$, compute the likelihood function $L(\beta | Y_1, \dots, Y_n, \Omega_1, \dots, \Omega_n)$ and give the estimating (score) equations for the maximum likelihood estimator of β .

Solution:

$$\begin{aligned} L(\beta | Y_1, \dots, Y_n, \Omega_1, \dots, \Omega_n) &= \prod_{i=1}^n \frac{(2\pi)^{n_i/2}}{|\Omega_i|^{1/2}} \exp(-(Y_i - X_i \beta)^T \Omega_i^{-1} (Y_i - X_i \beta)) \\ l(\beta | Y_1, \dots, Y_n, \Omega_1, \dots, \Omega_n) &= \sum_{i=1}^n \log\left(\frac{(2\pi)^{n_i/2}}{|\Omega_i|^{1/2}}\right) + (Y_i - X_i \beta)^T \Omega_i^{-1} (Y_i - X_i \beta) \\ \frac{\partial l}{\partial \beta}(\beta | Y_1, \dots, Y_n, \Omega_1, \dots, \Omega_n) &= \sum_{i=1}^n 2X_i^T \Omega_i^{-1} (Y_i - X_i \beta) \end{aligned}$$

Hence the estimating equation of the GLMM is

$$\sum_{i=1}^n X_i^T \Omega_i^{-1} (Y_i - X_i \beta) = 0.$$

We next consider a GEE specification with exchangeable correlation, that is

$$E[Y_{it}] = \mu_{it} = x_{it}^T \beta,$$

where we set $Var[Y_{it}] = \tau^2$ and the working correlation matrix $R_i(\alpha)$ such that its (t, s) element is equal to α , for $t \neq s$ and to 1 otherwise. We assume that τ^2 and α are known.

6. (3 points) Write explicitly the model estimating equations under this setting, based on the lecture slides (p. 190) and give the working covariance matrix $V_i = V_i(\alpha, \tau^2)$ explicitly.

Solution: Lecture slide 190 give that the estimating equation is :

$$\sum_{i=1}^n \left(\frac{\partial \mu_i}{\partial \beta} \right)^T V_i^{-1} (Y_i - \mu_i) = 0,$$

where $V_i = A_i^{1/2} R_i(\alpha) A_i^{1/2}$ with A_i a diagonal matrix with variances of $Y_{i1}, \dots, Y_{in_i}$ on its diagonal elements. Here, for any $t \in \{1, \dots, n_i\}$, $var(Y_{it}) = \tau^2$ such that $V_i = \tau^2 R_i(\alpha)$. Besides, $\mu_i = X_i \beta$ and $\frac{\partial \mu_i}{\partial \beta} = X_i$. Hence the estimating equation rewrites

$$\sum_{i=1}^n X_i^T \tau^{-2} R_i(\alpha)^{-1} (Y_i - \mu_i) = 0.$$

7. (2 points) Compare the estimating equations found in 5. and 6. What do you conclude for these two specifications under this specific setting? What is the relationship between σ_γ^2 , σ_ϵ^2 , τ^2 and α ? And between $R_i(\alpha)$ and Ω_i ?

Solution: Notice that the matrices Ω_i and $\tau^2 R_i(\alpha)$ have the same structure. Indeed, if

$$\begin{cases} \tau^2 &= \sigma_\epsilon^2 + \sigma_\gamma^2 \\ \alpha \tau^2 &= \sigma_\gamma^2 \end{cases} \iff \begin{cases} \tau^2 &= \sigma_\epsilon^2 + \sigma_\gamma^2 \\ \alpha &= \frac{\sigma_\gamma^2}{\sigma_\epsilon^2 + \sigma_\gamma^2} \end{cases},$$

$\Omega_i = \tau^2 V_i$ for all $i \in \{1, \dots, n\}$ and the estimating equations for β of the GEE and GLMM models become identical.

Output 8: GEE Model

```
1 > fit_gee <- gee(log(uptake) ~ log(conc-92) + Treatment + Type, id = Plant,
2   corstr = "exchangeable")
3
4 > summary(fit_gee)
5 GEE: GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
6 gee S-function, version 4.13 modified 98/01/27 (1998)
7
8 Model:
9 Link: Identity
10 Variance to Mean Relation: Gaussian
```

```

11 Correlation Structure:      Exchangeable
12
13 Call:
14 gee(formula = log(uptake) ~ log(conc-92) + Treatment + Type, id = Plant,
15 corstr = "exchangeable")
16
17 Summary of Residuals:
18 Min          1Q          Median          3Q          Max
19 -0.46055369 -0.09947886  0.01697381  0.12597427  0.33303499
20
21
22 Coefficients:
23 Estimate Naive S.E.   Naive z Robust S.E.   Robust z
24 (Intercept)      2.7084641 0.076396145 35.452890  0.09097667 29.770975
25 log(conc-92)      0.1773822 0.007700858 23.034085  0.01300256 13.642101
26 Treatmentchilled -0.2992991 0.075949456 -3.940767  0.07503523 -3.988780
27 TypeMississippi  -0.4895767 0.075949456 -6.446085  0.07503523 -6.524624
28
29 Estimated Scale Parameter:  0.0307418
30 Number of Iterations:  1
31
32 Working Correlation
33 [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
34 [1,] 1.0000000 0.4900651 0.4900651 0.4900651 0.4900651 0.4900651 0.4900651
35 [2,] 0.4900651 1.0000000 0.4900651 0.4900651 0.4900651 0.4900651 0.4900651
36 [3,] 0.4900651 0.4900651 1.0000000 0.4900651 0.4900651 0.4900651 0.4900651
37 [4,] 0.4900651 0.4900651 0.4900651 1.0000000 0.4900651 0.4900651 0.4900651
38 [5,] 0.4900651 0.4900651 0.4900651 0.4900651 1.0000000 0.4900651 0.4900651
39 [6,] 0.4900651 0.4900651 0.4900651 0.4900651 0.4900651 1.0000000 0.4900651
40 [7,] 0.4900651 0.4900651 0.4900651 0.4900651 0.4900651 0.4900651 1.0000000

```

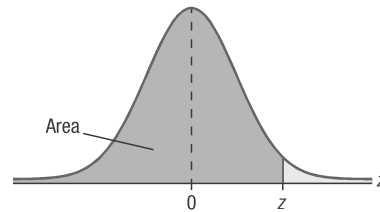
8. (1 point) We fitted a GEE model on the CO2 data. Output 8 shows a summary of this fit. Compare it to the GLMM Model summarized in Output 5 of the previous exercise. What do you notice ? Comment.

Solution: We observe that the regression parameter estimates ($\hat{\beta}$) are the same on both models, as expected. And this, even though the nuisance parameters are estimated.

B Standard Normal Distribution

Numerical entries represent the probability that a standard normal

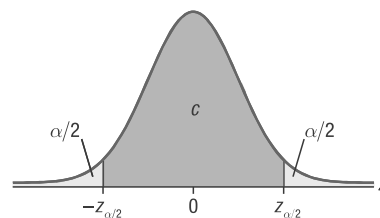
random variable is between $-\infty$ and z where $z = \frac{x - \mu}{\sigma}$.



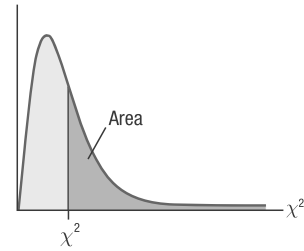
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Critical Values of z

Level of Confidence, c	$\alpha = 1 - c$	$z_{\alpha/2}$
0.80	0.20	1.28
0.85	0.15	1.44
0.90	0.10	1.645
0.95	0.05	1.96
0.98	0.02	2.33
0.99	0.01	2.575



H Critical Values of χ^2

Area to the Right of the Critical Value of χ^2

<i>df</i>	0.995	0.990	0.975	0.950	0.900	0.100	0.050	0.025	0.010	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169